

SYDNEY GRAMMAR SCHOOL



2017 Half-Yearly Examination

# FORM V

## MATHEMATICS EXTENSION 1

Tuesday 23rd May 2017

**General Instructions**

- Writing time — 1 hour 30 minutes
- Write using black pen.
- Board-approved calculators and templates may be used.

**Total — 80 Marks**

- All questions may be attempted.

**Section I – 8 Marks**

- Questions 1–8 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

**Section II – 72 Marks**

- Questions 9–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

**Collection**

- Write your name, class and Master on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your name, class and Master on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Nine.

5A: RCF

5B: SO

5C: BR

5D: REJ

5E: LYL

5F: LJF

5G: SDP

5H: CMDB

**Checklist**

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 152 boys

**Examiner**

LYL

## SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

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### QUESTION ONE

Given  $\alpha = \frac{-1 + \sqrt{7}}{2}$  and  $\beta = \frac{-1 - \sqrt{7}}{2}$ , what is the value of  $\alpha^2 - \beta^2$ ?

- (A) 1
- (B) 0
- (C)  $-\sqrt{7}$
- (D)  $-1$

### QUESTION TWO

Which of the following is a geometric sequence?

- (A) 2, 5, 11, 20, ...
- (B)  $8, -2, \frac{1}{2}, -\frac{1}{8}, \dots$
- (C)  $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \dots$
- (D) 22, 19, 16, 13, ...

### QUESTION THREE

Given that  $g(x) = x^2$ , what does  $g(5 - a)$  equal?

- (A)  $5 - a$
- (B)  $25 - a^2$
- (C)  $25 + 10a + a^2$
- (D)  $25 - 10a + a^2$

### QUESTION FOUR

What is the largest possible domain for the function  $f(x) = \frac{x}{\sqrt{9 - x^2}}$ ?

- (A)  $0 \leq x < 3$
- (B)  $-3 < x < 3$
- (C)  $0 \leq x < 9$
- (D)  $-9 < x < 9$

Examination continues next page ...

**QUESTION FIVE**

The point  $P$  divides the interval from  $A(-10, 8)$  to  $B(4, 1)$  internally in the ratio  $3 : 1$ .  
What are the coordinates of  $P$ ?

- (A)  $(\frac{1}{2}, \frac{11}{4})$
- (B)  $(-\frac{13}{2}, \frac{25}{4})$
- (C)  $(\frac{11}{4}, \frac{1}{2})$
- (D)  $(\frac{25}{4}, -\frac{13}{2})$

**QUESTION SIX**

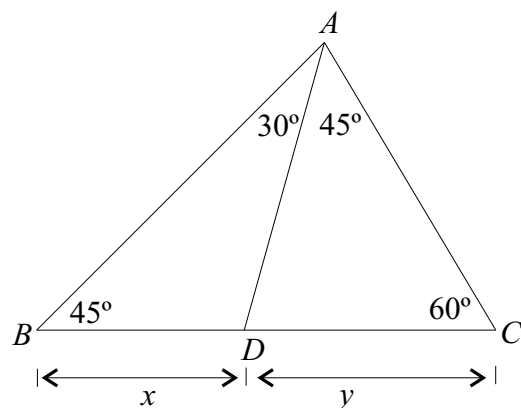
The derivative of  $f(x) = \frac{4}{x^3}$  is:

- (A)  $\frac{4}{3x^2}$
- (B)  $-\frac{12}{x^4}$
- (C)  $-\frac{12}{x^2}$
- (D)  $-\frac{4}{x^4}$

**Multiple choice continues on the next page**

**Examination continues overleaf ...**

**QUESTION SEVEN**



What is the value of  $\frac{x}{y}$  in the diagram above?

- (A)  $\frac{1}{\sqrt{3}}$
- (B)  $\sqrt{3}$
- (C)  $\frac{\sqrt{3}}{2}$
- (D)  $\frac{2}{\sqrt{3}}$

**QUESTION EIGHT**

Consider the equation  $\tan(2x - 45) = -1$ , for  $0^\circ \leq x \leq 360^\circ$ . How many solutions does this equation have in this domain?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

\_\_\_\_\_ End of Section I \_\_\_\_\_

**Examination continues next page ...**

**SECTION II - Written Response**

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

**QUESTION NINE** (12 marks) Use a separate writing booklet.

**Marks**

(a) Simplify  $(4\sqrt{5} - 1)^2$ . **1**

(b) Differentiate:

(i)  $y = 2x^3 - 3x$  **1**

(ii)  $y = x(x + 3)$  **1**

(iii)  $y = \frac{x^3 + x}{x}$  **1**

(c) Solve  $|2x - 7| > 3$ . **2**

(d) Consider the sequence 40, 38, 36, ... .

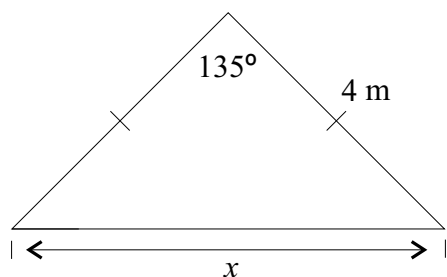
(i) State the common difference. **1**

(ii) Find the thirteenth term. **1**

(e) Factorise  $27 - x^3$ . **1**

(f) Evaluate  $\sum_{n=1}^5 (n^3 - 1)$ . **1**

(g)



The diagram above shows the gable of a roof. The gable forms an isosceles triangle with two rafters of 4 metres and an angle of  $135^\circ$  at the apex.

(i) Find the value of  $x$ . Give your answer correct to two decimal places. **1**

(ii) Find the exact area of the gable. **1**

**Examination continues overleaf ...**

**QUESTION TEN** (12 marks) Use a separate writing booklet. **Marks**

- (a) Find the perpendicular distance between the parallel lines  $3x + 2y - 1 = 0$  and  $3x + 2y + 4 = 0$ . **2**
- (b) Solve  $4 \cos^2 \theta = 3$ , for  $0^\circ \leq \theta \leq 360^\circ$ . **2**
- (c) Sketch  $y = \frac{1}{x+4}$ . Show all asymptotes and any intercepts with the axes. **2**
- (d) Find the equation of the tangent to the curve  $y = 3x^2 + x - 6$  at the point  $(-1, -4)$ . Give your answer in gradient–intercept form. **2**
- (e) Solve the inequation  $\frac{4}{x-1} \leq 2$ . **4**

**QUESTION ELEVEN** (12 marks) Use a separate writing booklet. **Marks**

- (a) Differentiate:
- (i)  $y = (4x - 1)^6$  **2**
- (ii)  $y = x(3x + 2)^4$  **2**
- (b) Given  $f(x) = \sqrt{4 - x^2}$ , sketch  $y = f\left(\frac{1}{2}x\right)$  and clearly mark any significant features. **2**
- (c) Prove  $\frac{\sin \theta + \cos \theta}{\cot \theta + 1} = \sin \theta$ . **2**
- (d) Consider the graphs of  $y = 6x - x^2$  and  $y = 2x$ .
- (i) Find any points of intersection. **1**
- (ii) Shade the intersection of the regions  $y \leq 6x - x^2$  and  $y > 2x$ . Indicate the nature of the boundaries and corners clearly. **3**

**QUESTION TWELVE** (12 marks) Use a separate writing booklet.

**Marks**

- (a) If  $\cos \theta = -\frac{4}{7}$  and  $\tan \theta > 0$ , find the exact value of  $\sin \theta$ . **2**
- (b) The third term of a geometric sequence is 54 and the sixth term is 2. Find the common ratio. **1**
- (c) Determine whether  $f(x) = \frac{x^2 - 3}{x}$  is even, odd or neither. Show your working clearly. **2**
- (d) Find the angle of inclination of the line  $2x + 3y - 1 = 0$ . Give your answer correct to the nearest minute. **2**
- (e) Given  $\log_b z = \log_b(x + 2) - 2\log_b y$ , write an expression for  $z$  without logarithms. **2**
- (f) Consider the function  $f(x) = x^2 - 5x$ .
- (i) Simplify the expression  $f(x + h) - f(x)$ . **2**
- (ii) Hence use the formula **1**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

to differentiate  $f(x) = x^2 - 5x$  from first principles.

**QUESTION THIRTEEN** (12 marks) Use a separate writing booklet.

**Marks**

- (a) Without finding the point of intersection, find the equation of the line that passes through  $2x - y + 1 = 0$  and  $x + 3y - 5 = 0$  and is parallel to the line  $y = -2x + 3$ . Give your answer in general form. **3**
- (b) Solve  $\tan^2 x - 2\sec^2 x + 3 = 0$ , for  $0^\circ \leq x \leq 180^\circ$ . **3**
- (c) Consider the graph  $y = \frac{x - 1}{x^2 - 2x - 3}$ .
- (i) Find any intercepts with the coordinate axes. **1**
- (ii) Write down the equation(s) of any vertical asymptote(s). **1**
- (iii) Find the equation of the horizontal asymptote. **1**
- (iv) Copy and complete the following table. **1**

|     |    |   |   |   |
|-----|----|---|---|---|
| $x$ | -2 | 0 | 2 | 4 |
| $y$ |    |   |   |   |

- (v) Draw a neat sketch of the graph. Show all the features from parts (i) to (iv). **2**

**Examination continues overleaf ...**

**QUESTION FOURTEEN** (12 marks) Use a separate writing booklet.

**Marks**

(a) Factorise fully  $x^4 - 3x^2y^2 - 4y^4$ .

**1**

(b) A function is defined by the rule:

**2**

$$f(x) = \begin{cases} (x-1)^3 & \text{for } x \geq 0 \\ (1-x)^3 & \text{for } x < 0 \end{cases}$$

Find a simplified expression for  $f\left(\frac{a^2}{2}\right) + f\left(\frac{-a^2}{2}\right)$ .

(c) (i) Differentiate  $y = \frac{2x+1}{\sqrt{2x-1}}$ .

**2**

(ii) Find the  $x$ -coordinate of any point(s) where the tangent is horizontal.

**1**

(d) It is known that  $x, y, z$  are the first three terms of a geometric progression and  $x, z, y$  are the first three terms of an arithmetic progression.

(i) Show that  $2\left(\frac{y}{x}\right)^2 - \left(\frac{y}{x}\right) - 1 = 0$ .

**2**

(ii) Given the geometric sequence has a limiting sum, find  $S_\infty$  in terms of  $x$ .

**2**

(e) Consider the function  $f(x) = ||x+3| - 1|$ . For what values of the constant  $k$  does the equation  $f(x) = k$  have exactly four distinct solutions?

**2**

\_\_\_\_\_ End of Section II \_\_\_\_\_

**END OF EXAMINATION**

NAME: .....

CLASS: ..... MASTER: .....

SYDNEY GRAMMAR SCHOOL



2017

Half-Yearly Examination

FORM V

MATHEMATICS EXTENSION 1

Tuesday 23rd May 2017

- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

**Question One**

A ☐ B ☐ C ☐ D ☐

**Question Two**

A ☐ B ☐ C ☐ D ☐

**Question Three**

A ☐ B ☐ C ☐ D ☐

**Question Four**

A ☐ B ☐ C ☐ D ☐

**Question Five**

A ☐ B ☐ C ☐ D ☐

**Question Six**

A ☐ B ☐ C ☐ D ☐

**Question Seven**

A ☐ B ☐ C ☐ D ☐

**Question Eight**

A ☐ B ☐ C ☐ D ☐

# Form V Extension 1 Half yearly Solutions

Ext 1 Half yearly 2017  
Multiple choice

Q1 (C)

$$\alpha = \frac{-1 + \sqrt{7}}{2}$$

$$= \frac{\sqrt{7} - 1}{2}$$

$$\alpha^2 = \frac{(\sqrt{7} - 1)^2}{4}$$

$$= \frac{7 - 2\sqrt{7} + 1}{4}$$

$$= \frac{8 - 2\sqrt{7}}{4}$$

$$= \frac{2(4 - \sqrt{7})}{4}$$

$$= \frac{4 - \sqrt{7}}{2}$$

$$\beta = \frac{-1 - \sqrt{7}}{2}$$

$$= -\frac{(\sqrt{7} + 1)}{2}$$

$$\beta^2 = \frac{(\sqrt{7} + 1)^2}{4}$$

$$= \frac{7 + 2\sqrt{7} + 1}{4}$$

$$= \frac{2(4 + \sqrt{7})}{4}$$

$$= \frac{4 + \sqrt{7}}{2}$$

$$\alpha^2 - \beta^2 = \frac{4 - \sqrt{7}}{2} - \frac{4 + \sqrt{7}}{2}$$

$$= \frac{-2\sqrt{7}}{2}$$

$$= -\sqrt{7}$$

Q2 (B)

$$\text{If } \frac{T_2}{T_1} = \frac{T_3}{T_2}$$

$$\text{or } T_2^2 = T_1 \times T_3$$

} then GP

Sequence  $8, -2, \frac{1}{2}, -\frac{1}{8}, \dots$   
 $T_1, T_2, T_3$

$$T_1 \times T_3 = 8 \times \frac{1}{2}$$

$$= 4$$

$$T_2^2 = (-2)^2$$

$$= 4$$

$\therefore 8, -2, \frac{1}{2}, -\frac{1}{8}, \dots$  is a GP.

Q3

$$g(x) = x^2$$

$$g(5-a) = (5-a)^2$$

$$= 25 - 10a + a^2$$

(D)

Q4

$$f(x) = \frac{x}{\sqrt{9-x^2}}$$

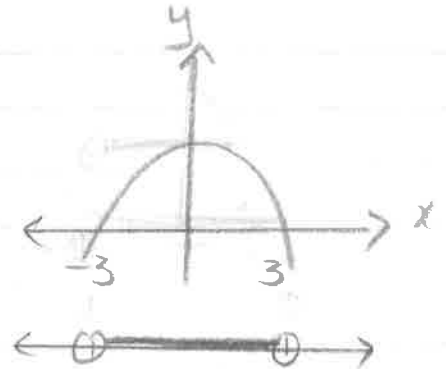
Domain

$$\sqrt{9-x^2} > 0$$

$$9-x^2 > 0$$

$$(3-x)(3+x) > 0$$

$$-3 < x < 3$$



Q5

$$A(-10, 8)$$

$$x_1, y_1$$

$$B(4, 1)$$

$$x_2, y_2$$

$$3:1$$

$$k:l$$

$$xL = \frac{kx_2 + lx_1}{k+l}$$

$$= \frac{3 \times 4 + 1 \times -10}{4}$$

$$= \frac{12 - 10}{4}$$

$$= \frac{1}{2}$$

$$y = \frac{ky_2 + ly_1}{k+l}$$

$$= \frac{3 \times 1 + 1 \times 8}{4}$$

$$= \frac{11}{4}$$

$$M\left(\frac{1}{2}, \frac{11}{4}\right)$$

(A)

Q6

$$f(x) = 4x^{-3}$$

$$f'(x) = -12x^{-4}$$

$$= -\frac{12}{x^4}$$

(B)

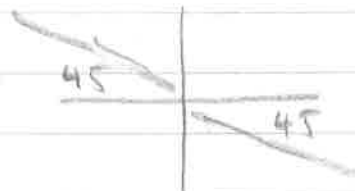
Q7  $\tan(2x - 45) = -1$   
 let  $\theta = 2x - 45$

$0^\circ < x \leq 360^\circ$   
 $0^\circ < 2x \leq 720^\circ$   
 $-45 \quad -45 \quad -45$

$\tan \theta = -1$

related angle  $= 45^\circ$

$\theta = -45^\circ, 135^\circ, 315^\circ, 495^\circ, 675^\circ$



Since  $\theta = 2x - 45$

$x = \frac{\theta + 45}{2}$

(D)

$x = 0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$

$\therefore$  There are 5 solutions.

Q8 In  $\triangle ADB$

$\frac{x}{\sin 30^\circ} = \frac{AD}{\sin 45^\circ}$

$x = \frac{AD \sin 30^\circ}{\sin 45^\circ}$

In  $\triangle ADC$

$\frac{y}{\sin 45^\circ} = \frac{AD}{\sin 60^\circ}$

$y = \frac{AD \sin 45^\circ}{\sin 60^\circ}$

$\frac{x}{y} = \frac{AD \sin 30^\circ}{\sin 45^\circ} \div \frac{AD \sin 45^\circ}{\sin 60^\circ}$

$= \frac{\cancel{AD} \sin 30^\circ}{\sin 45^\circ} \times \frac{\sin 60^\circ}{\cancel{AD} \sin 45^\circ}$

$= \frac{1}{2} \div \frac{1}{\sqrt{2}} \times \frac{\frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}}$

$= \frac{\cancel{\sqrt{2}}}{2} \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{1}$

$= \frac{\sqrt{3}}{2}$

(C)

Q9 a)  $(4\sqrt{5}-1)^2$   
 $= (4\sqrt{5})^2 - 2 \cdot 4\sqrt{5} + 1$   
 $= 16 \times 5 - 8\sqrt{5} + 1$   
 $= \underline{81 - 8\sqrt{5}}$  ✓

b) i)  $y = 2x^3 - 3x$   
 $\underline{y' = 6x^2 - 3}$  ✓

ii)  $y = x(x+3)$   
 $= x^2 + 3x$   
 $\underline{y' = 2x + 3}$  ✓

iii)  $y = \frac{x^3 + x}{x}$   
 $= x^2 + 1$   
 $\underline{y' = 2x}$  ✓

c)  $|2x - 7| > 3$   
 $2x - 7 > 3$   
 $2x > 10$   
 $\underline{x > 5}$  ✓

$2x - 7 < -3$   
 $2x < 4$   
 $\underline{x < 2}$  ✓

d) i)  $\underline{d = -2}$  ✓  
 $a = 40$

ii)  $T_n = a + (n-1)d$   
 $T_{13} = 40 + 12 \times -2$   
 $= 40 - 24$   
 $\underline{T_{13} = 16}$  ✓

e)  $27 - x^3 = 3^3 - x^3$   
 $= \underline{(3-x)(9 + 3x + x^2)}$  ✓

Q9 continued

$$\begin{aligned} f) \sum_{n=1}^5 (n^3 - 1) &= 0 + (2^3 - 1) + (3^3 - 1) + (4^3 - 1) + (5^3 - 1) \\ &= 8 + 27 + 64 + 125 - 4 \\ &= 224 - 4 \\ &= \underline{220} \quad \checkmark \end{aligned}$$

$$\begin{aligned} g) \text{ i) } x^2 &= 4^2 + 4^2 - 2 \cdot 4 \cdot 4 \cdot \cos 135^\circ \\ &= 32 - 2 \cdot 4 \cdot 4 \times \frac{1}{\sqrt{2}} \\ &= 32 - \cancel{2} \cdot 4 \cdot 4 \times \frac{\sqrt{2}}{\cancel{2}} \\ &= 32 - 16\sqrt{2} \\ x &= \sqrt{32 - 16\sqrt{2}} \\ x &\doteq \underline{7.39\text{m}} \quad (2\text{dp}) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{ii) Area} &= \frac{1}{2} ab \sin 135^\circ \\ &= \frac{1}{2} \times 16 \times \frac{1}{\sqrt{2}} \\ &= \frac{8}{\sqrt{2}} \times \sqrt{2} \\ &= \underline{4\sqrt{2} \text{ u}^2} \quad \checkmark \end{aligned}$$

Q10 a)  $3x + 2y - 1 = 0$

Sub  $x = 1$

$$3 + 2y - 1 = 0$$

$$2y + 2 = 0$$

$$y = -1$$

$$(1, -1) \checkmark$$

accept  
any point on  
either line

Perpendicular distance

$(1, -1)$  to  $3x + 2y + 4 = 0$

$$a = 3$$

$$b = 2$$

$$c = 4$$

$$P = \frac{|3 \times 1 + 2 \times -1 + 4|}{\sqrt{3^2 + 2^2}}$$

$$\sqrt{3^2 + 2^2}$$

$$= \frac{5}{\sqrt{13}} \checkmark$$

$$= \frac{5\sqrt{13}}{13} \text{ units}$$

b)

$$4 \cos^2 \theta = 3$$

$$\cos^2 \theta = \frac{3}{4}$$

$$\cos \theta = \pm \frac{\sqrt{3}}{2} \checkmark$$

$$\cos \theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ, 330^\circ$$

$$\cos \theta = -\frac{\sqrt{3}}{2}$$

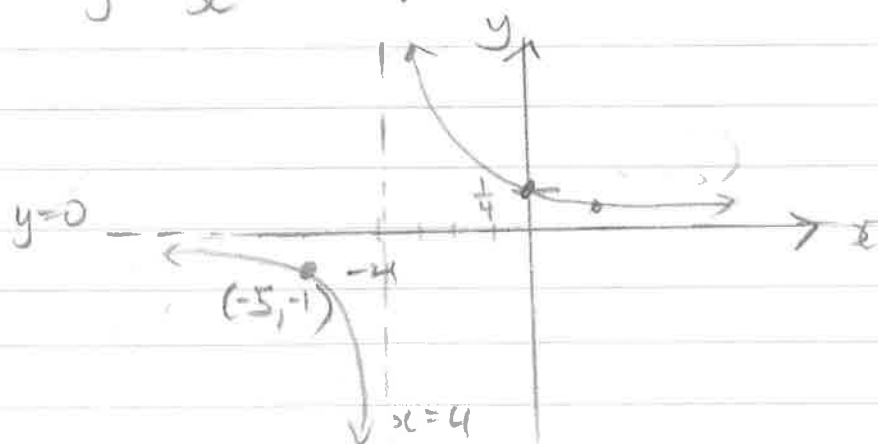
$$\theta = 150^\circ, 210^\circ$$

$\checkmark$

Q10 continued

c)  $y = \frac{1}{x+4}$

$y = \frac{1}{x}$  shifted left 4 units.



y-intercept  
 $x=0 \quad y = \frac{1}{4}$

asymptotes  
vertical  $x = -4$   
horizontal  
 $y = 0$

|   |                |    |               |               |
|---|----------------|----|---------------|---------------|
| x | -6             | -5 | 0             | 1             |
| y | $-\frac{1}{2}$ | -1 | $\frac{1}{4}$ | $\frac{1}{5}$ |

✓ shape

d)  $y = 3x^2 + x - 6$   
 $y' = 6x + 1$

At  $(-1, -4) \quad y' = -6 + 1 = -5$

$y - y_1 = m(x - x_1)$   
 $y + 4 = -5(x + 1)$   
 $y = -5x - 9$

e)  $\frac{4}{x-1} \leq 2$

\*  $x \neq 1$

$(x-1) \times \frac{4}{x-1} \leq 2(x-1)^2$  ✓

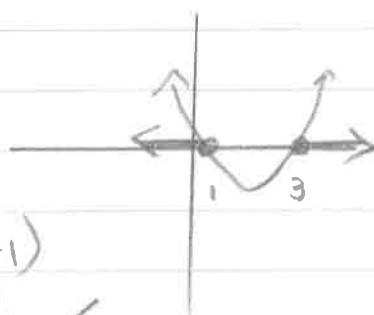
$4(x-1) \leq 2(x-1)^2$

$0 \leq 2(x-1)^2 - 4(x-1)$

$0 \leq 2(x-1)[(x-1)-2]$  ✓

$2(x-1)(x-3) \geq 0$

$x < 1$  ✓ or  $x \geq 3$  ✓



Q11 a) i)  $y = (4x-1)^6$  ✓  
 $y' = 6(4x-1)^5 \cdot 4$  ✓  
 $= 24(4x-1)^5$

Chain rule

ii)  $y = x(3x+2)^5$   
 $y = uv$

$u = x$   
 $u' = 1$

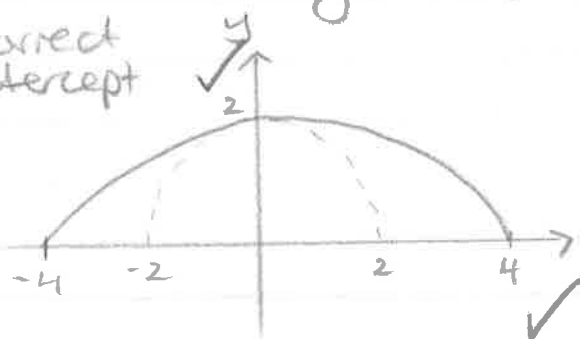
$v = (3x+2)^5$   
 $v' = 5(3x+2)^4 \cdot 3$   
 $= 15(3x+2)^4$  ✓

$y' = v u' + u v'$   
 $= (3x+2)^4 + 15x(3x+2)^4$  ✓  
 $= (3x+2)^4 [1 + 15x]$   
 $= (3x+2)^4 (15x+1)$

b)  $f(x) = \sqrt{4-x^2}$   
 positive semi-circle radius 2

$f(\frac{1}{2}x)$  stretched horizontally  
 by a factor of 2

correct  
y-intercept



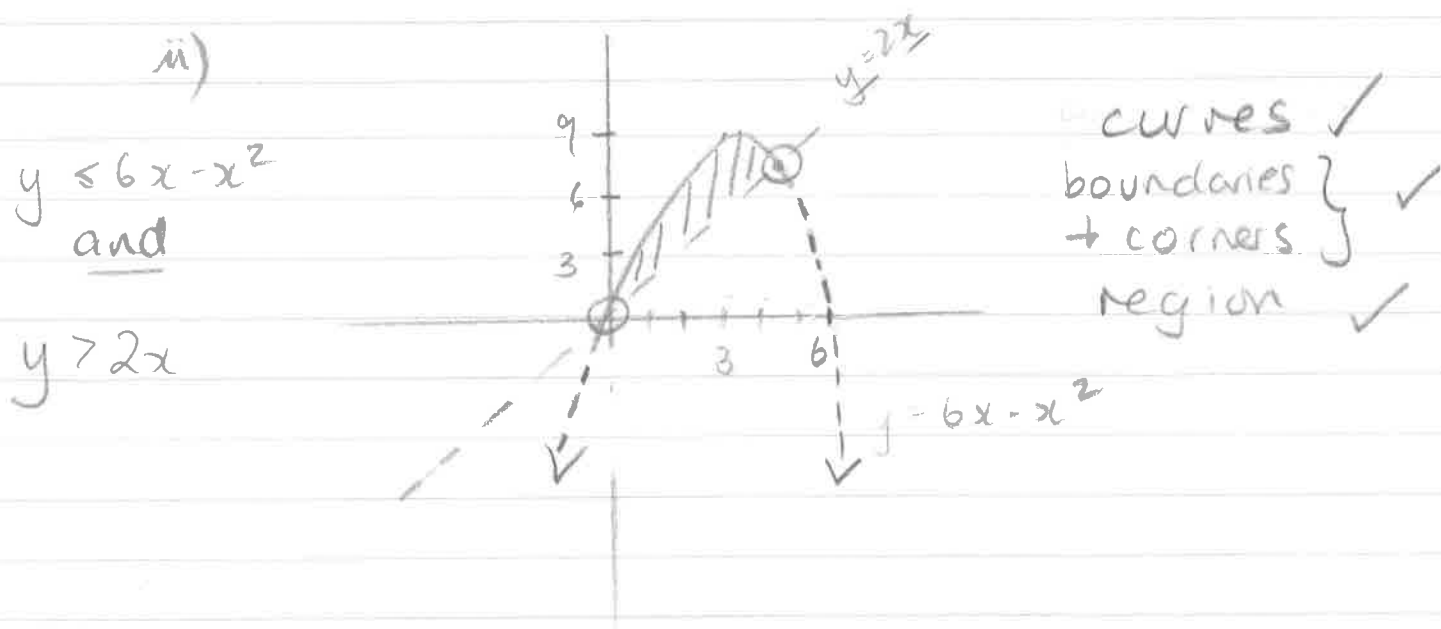
let  $y = f(x)$   
 $y = \sqrt{4-x^2}$

correct x intercepts

$$\begin{aligned}
 c) \text{ LHS} &= \frac{\sin \theta + \cos \theta}{\cot \theta + 1} \\
 &= \frac{\sin \theta + \cos \theta}{\frac{\cos \theta}{\sin \theta} + 1} \quad \checkmark \\
 &= \frac{(\sin \theta + \cos \theta)}{1} \div \frac{\cos \theta + \sin \theta}{\sin \theta} \\
 &= \frac{\cancel{\sin \theta + \cos \theta}}{1} \times \frac{\sin \theta}{\cancel{\cos \theta + \sin \theta}} \quad \checkmark \\
 &= \sin \theta \\
 &= \text{RHS as required.}
 \end{aligned}$$

$$\begin{aligned}
 d) \text{ i) } y &= 6x - x^2 & y &= 2x \\
 6x - x^2 &= 2x \\
 4x - x^2 &= 0 \\
 x(4 - x) &= 0 \\
 x &= 0 & \text{or } x &= 4
 \end{aligned}$$

Points of intersection  
(0, 0) and (4, 8) ✓



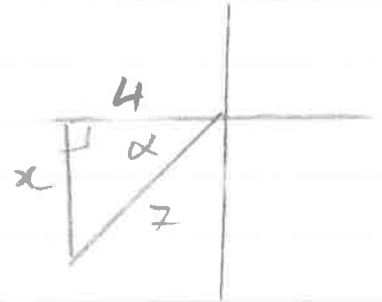
Q12) a)  $\cos \theta = -\frac{4}{7}$   $\tan \theta > 0$

$$x^2 + 4^2 = 7^2$$

$$x^2 = 49 - 16$$

$$= 33$$

$$x = \sqrt{33} \quad (x > 0) \quad \checkmark$$



$$\sin \theta = -\frac{\sqrt{33}}{7} \quad \checkmark$$

b) GP  $T_n = ar^{n-1}$

$$T_3 = 54$$

$$ar^2 = 54 \quad (1)$$

$$T_6 = 2$$

$$ar^5 = 2 \quad (2)$$

$$(2) \div (1)$$

$$\frac{ar^5}{ar^2} = \frac{2}{54}$$

$$r^3 = \frac{1}{27}$$

$$r = \frac{1}{3} \quad \checkmark$$

c)  $f(x) = \frac{x^2 - 3}{x}$

$$f(-x) = \frac{(-x)^2 - 3}{-x} \quad \checkmark$$

$$= \frac{x^2 - 3}{-x}$$

$$= -f(x) \quad \checkmark$$

$\therefore f(x)$  is odd.

Q 12 continued

$$\begin{aligned} \text{d) } 2x + 3y - 1 &= 0 \\ 3y &= -2x + 1 \\ y &= -\frac{2}{3}x + \frac{1}{3} \end{aligned}$$

$$\begin{aligned} m &= -\frac{2}{3} \checkmark \\ \tan \theta &= -\frac{2}{3} \end{aligned}$$



$$\tan \alpha = \frac{2}{3}$$

$$\begin{aligned} \theta &\doteq 180 - 33^\circ 41' \checkmark \\ &\doteq 146^\circ 19' \text{ (nearest minute)} \end{aligned}$$

$$\text{e) } \log_b z = \log_b \left( \frac{x+z}{y^2} \right) \checkmark$$

$$z = \frac{x+z}{y^2} \checkmark$$

$$\text{f) } f(x) = x^2 - 5x$$

$$\begin{aligned} f(x+h) &= (x+h)^2 - 5(x+h) \\ &= x^2 + 2hx + h^2 - 5x - 5h \checkmark \end{aligned}$$

$$\begin{aligned} f(x+h) - f(x) &= x^2 + 2hx + h^2 - 5x - 5h \\ &\quad - x^2 \quad \quad \quad + 5x \\ &= 2hx + h^2 - 5h \\ &= h(2x + h - 5) \checkmark \end{aligned}$$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \cancel{h} \frac{(2x + h - 5)}{\cancel{h}} \\ &= \underline{2x - 5} \checkmark \end{aligned}$$

Q13

$$a) \quad 2x - y + 1 + k(x + 3y - 5) = 0 \quad *$$

$$(2x + kx)(-y + 3ky) + (1 - 5k) = 0$$

$$(2+k)x + (3k-1)y + (1-5k) = 0$$

$$(3k-1)y = -(2+k)x - (1-5k) \quad \checkmark$$

$$y = \frac{-(2+k)}{(3k-1)}x - \frac{(1-5k)}{(3k-1)}$$

$$\frac{2+k}{1-3k} = -2$$

$$1-3k$$

$$2+k = -2(1-3k)$$

$$2+k = -2 + 6k$$

$$4 = 5k$$

$$k = \frac{4}{5} \quad \checkmark$$

Sub  $k = \frac{4}{5}$  into \*

$$2x - y + 1 + \frac{4}{5}(x + 3y - 5) = 0$$

$$5(2x - y + 1) + 4x + 12y - 20 = 0$$

$$\underline{10x} - \underline{5y} + 5 + \underline{4x} + \underline{12y} - 20 = 0$$

$$\underline{14x + 7y - 15 = 0} \quad \checkmark$$

asked for general form

Q13 continued

$$b) \tan^2 x - 2 \sec^2 x + 3 = 0$$

$$\underline{0^\circ \leq x \leq 180^\circ}$$

$$\tan^2 x - 2(1 + \tan^2 x) + 3 = 0$$

$$\tan^2 x - 2 - 2\tan^2 x + 3 = 0$$

$$-\tan^2 x + 1 = 0$$

$$\tan^2 x - 1 = 0$$

$$(\tan x - 1)(\tan x + 1) = 0$$

$$\tan x - 1 = 0$$

$$\tan x = 1$$

$$x = 45^\circ, 225^\circ$$

but domain

$$0^\circ \leq x \leq 180^\circ$$

$$\therefore x = 45^\circ \text{ or } 135^\circ$$

$$\tan x + 1 = 0$$

$$\tan x = -1$$

$$x = 135^\circ, 315^\circ$$

✓

Q13 continued

c)  $y = \frac{x-1}{x^2-2x-3}$

i)  $x=0 \quad y = \frac{1}{3}$

$y=0 \quad x-1=0$   
 $x=1$

$(0, \frac{1}{3})$  y-intercept  
accept  $y = \frac{1}{3}$   
 $(1, 0)$  x-intercept

ii) vertical asymptotes  
 $x = -1$  or  $x = 3$

iii)  $y = \frac{\frac{x}{x^2} - \frac{1}{x}}{\frac{x^2}{x^2} - \frac{2x}{x^2} - \frac{3}{x^2}}$   
 $= \frac{0}{1 - \frac{2}{x} - \frac{3}{x^2}}$

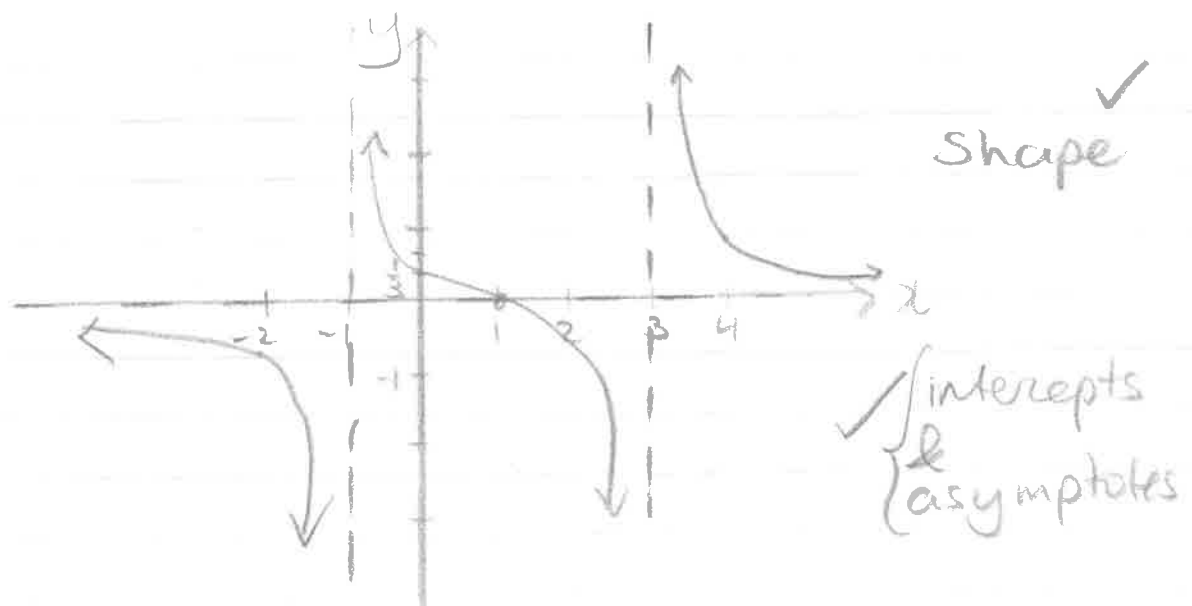
$x \rightarrow \infty \quad y \rightarrow 0$   
 $x \rightarrow -\infty \quad y \rightarrow 0$

$y=0$   
horizontal asymptote

iv)

|   |                |               |                |               |
|---|----------------|---------------|----------------|---------------|
| x | -2             | 0             | 2              | 4             |
| y | $-\frac{3}{5}$ | $\frac{1}{3}$ | $-\frac{1}{3}$ | $\frac{3}{5}$ |

v)



$$814 \text{ a) } x^4 - 3x^2y^2 - 4y^4 = (x^2 - 4y^2)(x^2 + y^2) \\ = (x - 2y)(x + 2y)(x^2 + y^2) \checkmark$$

$$\text{b) } f(x) = \begin{cases} (x-1)^3 & \text{for } x \geq 0 \\ (1-x)^3 & \text{for } x < 0 \end{cases}$$

$$f\left(\frac{a^2}{2}\right) = \left(\frac{a^2}{2} - 1\right)^3$$

$$f\left(-\frac{a^2}{2}\right) = \left(1 - \left(-\frac{a^2}{2}\right)\right)^3 \\ = \left(\frac{a^2}{2} + 1\right)^3$$

$$\left\{ \begin{array}{l} \text{Special case} \\ \text{when } a = 0 \\ f\left(\frac{a^2}{2}\right) = -1 \\ f\left(-\frac{a^2}{2}\right) = -1 \\ \therefore f\left(\frac{a^2}{2}\right) + f\left(-\frac{a^2}{2}\right) = -2 \end{array} \right.$$

$$f\left(\frac{a^2}{2}\right) + f\left(-\frac{a^2}{2}\right)$$

$$= \left(\frac{a^2}{2} - 1\right)^3 + \left(\frac{a^2}{2} + 1\right)^3 \quad \checkmark^*$$

$$= \left(\frac{a^2}{2} - 1 + \frac{a^2}{2} + 1\right) \left[ \left(\frac{a^2}{2} - 1\right)^2 - \left(\frac{a^2}{2} - 1\right)\left(\frac{a^2}{2} + 1\right) + \left(\frac{a^2}{2} + 1\right)^2 \right]$$

$$= a^2 \left[ \left(\frac{a^4}{4} - \frac{2a^2}{2} + 1\right) - \left(\frac{a^4}{4} - 1\right) + \left(\frac{a^4}{4} + \frac{2a^2}{2} + 1\right) \right]$$

$$= a^2 \left( 3 + \frac{a^4}{4} \right) \quad \checkmark$$

Accept expansion of  $f\left(\frac{a^2}{2}\right)$  and  $f\left(-\frac{a^2}{2}\right)$

$$f\left(\frac{a^2}{2}\right) = \left(\frac{a^2}{2} - 1\right)^3$$

$$= \frac{a^6}{8} - \frac{3a^4}{4} + \frac{3a^2}{2} + 1$$

$$f\left(-\frac{a^2}{2}\right) = \left(\frac{a^2}{2} + 1\right)^3$$

$$= \frac{a^6}{8} + \frac{3a^4}{4} + \frac{3a^2}{2} + 1$$

$\left. \begin{array}{l} \text{Alternative} \\ \text{method} \\ 1 \text{ mark} \\ \text{for} \\ \text{correct} \\ \text{expansion} \end{array} \right\}$

Q14 continued

i)  $y = \frac{2x+1}{\sqrt{2x-1}}$

let  $y = \frac{u}{v}$

$$u = 2x+1$$

$$u' = 2$$

$$v = (2x-1)^{\frac{1}{2}}$$

$$v' = \frac{1}{2}(2x-1)^{-\frac{1}{2}} \cdot 2$$

$$= (2x-1)^{-\frac{1}{2}}$$
$$= \frac{1}{\sqrt{2x-1}}$$

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{2\sqrt{2x-1} - \frac{(2x+1)}{\sqrt{2x-1}}}{2x-1} \quad \checkmark$$

$$= \frac{2(2x-1) - (2x+1)}{\sqrt{2x-1}} \times \frac{1}{2x-1}$$

$$= \frac{4x-2-2x-1}{(2x-1)^{\frac{3}{2}}}$$

$$= \frac{2x-3}{(2x-1)^{\frac{3}{2}}} \quad \checkmark$$

ii) when  $y' = 0$

$$2x-3=0$$

$$2x=3$$

$$x = \frac{3}{2} \quad \checkmark$$

Q14 d) GP  $x, y, z$

so

$$\frac{y}{x} = \frac{z}{y}$$

$$r = \frac{y}{x}$$

$$z = \frac{y^2}{x} \quad (1)$$

AP  $x, z, y$

$$z - x = y - z$$

$$2z = x + y \quad (2)$$

✓ two equations

Sub (1) into (2)

$$2 \frac{y^2}{x} = x + y$$

$$2y^2 = x^2 + xy$$

$$\div x^2 \quad 2y^2 - xy - x^2 = 0$$

$$2 \frac{y^2}{x^2} - \frac{xy}{x^2} - \frac{x^2}{x^2} = 0$$

$$2 \left( \frac{y}{x} \right)^2 - \left( \frac{y}{x} \right) - 1 = 0 \quad \text{as required}$$

GP has a limiting sum

$$\therefore |r| < 1 \quad \left| \frac{y}{x} \right| < 1$$

$$\text{let } r = \frac{y}{x}$$

$$2r^2 - r - 1 = 0$$

$$(2r+1)(r-1) = 0$$

$$2r+1=0$$

$$2r=-1$$

$$r = -\frac{1}{2}$$

$$r-1=0$$

$$r=1$$

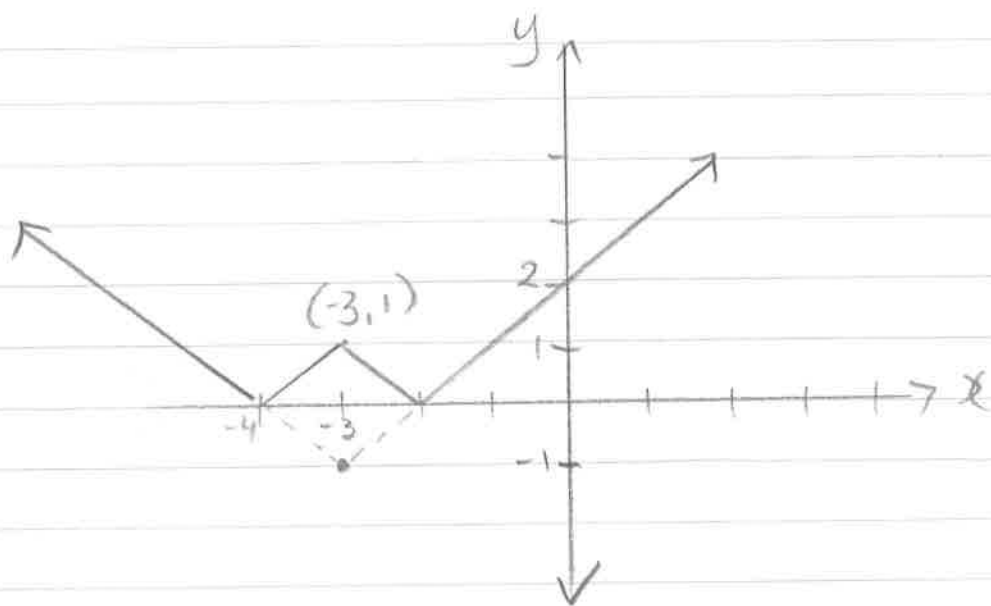
$$|r| < 1 \quad \therefore r = -\frac{1}{2}$$

$$S_{\infty} = \frac{a}{1-r} = \frac{x}{\frac{3}{2}} = \frac{2x}{3}$$

Q14 e)  $f(x) = ||x+3|-1|$   
 let  $y = f(x)$ .

Standard curve  $y = |x|$  is  
 shifted left 3 units

then  $y = |x+3|$  is shifted down 1 unit  
 and then curve below x-axis  
 is reflected in x-axis.



$||x+3|-1| = k$  exactly four solutions.  
 $0 < k < 1$  ✓